INVESTIGATING GPS ACCURACY USING NOISE REDUCTION FOR LAUNCH APPLICATIONS USING R

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Abstract:

In the past decades, astronautic engineers have found ways to safely land rockets on planets through remarkable technology: the Global Positioning System, otherwise known as GPS. The Global Positioning System consists of 29 revolving around our planet, constantly sending out signals which then are intercepted by receivers (e.x. iphone) to locate your current position. This research investigates the evolution of GPS and how it has improved over years. GPS helps mission control track and communicate with rockets more reliably, potentially reducing risks during launch or orbit insertion phases. However, GPS is not always reliable - sometimes it gets interfered with atmospheric disturbances, signal multipaths, and other artifacts. This research also dives further about data in which different interruptions affect rocket launches the most.\

In the end, however, there are many ways to improve GPS tracking systems for better rocket launches. We can apply a suite of noise-removal techniques, including moving-average filtering, wavelet denoising, and the Kolmogorov–Zurbenko (KZ) filter to suppress short-term atmospheric and multipath disturbances, then compare positional accuracy before and after cleaning. Finally, we discuss how these enhanced signals can improve mission-control tracking and reduce launch-phase risks during orbit insertion.

Background Information:

- The Global Positioning System (GPS) was operational by United States Department of Defense 1995
- The GPS provides precise destination, velocity, and timing data to help with rocket launches.
- GPS systems aren't perfectly accurate, it has interruptions from the atmosphere.
- Our goal is to improve accuracy of GPS systems by removing outliers.

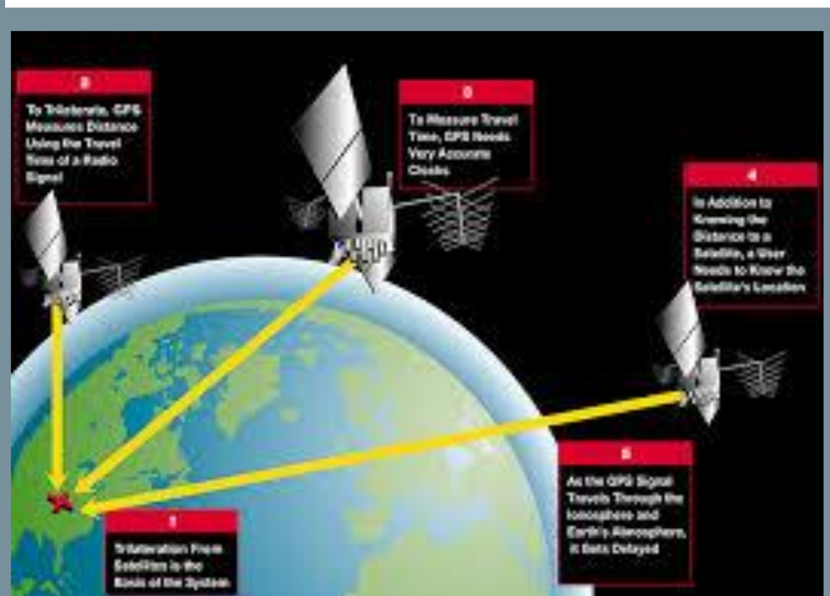


Figure 1: Conceptual figure showing the steps of the Global Positioning System (GPS) in its effect for rocket launches.

Method/Materials:

- **Canva**
  - Creating examples of information using data gathered from outside resources.
- **Nevada Geodetic Laboratory**
  - Processed GPS data from various systems in Long Island
  - A map of Long Island's GPS stations.
- **Global Mapper**
  - Creation of a path profile of Long Island
  - Shows sites of landslides have higher elevation
- **RStudio (v.4.2.1)**
  - Created examples of cleaned and raw data using packages
    The cleaned data provided smoother, more accurate position trends useful for assessing GPS
  - Included noise reduction using zoo package

Results:

- After using RStudio to compare graphs of raw vs cleaned data of GPS outliers, we can conclude that cleaning reduces spikes and noise, resulting in an improvement of positional accuracy (see Figure 4)

- Accuracy can be further improved by:

- topographic location due to natural disturbances,
- using correction data from ground stations,
- and enhancing onboard receiver hardware

These improvements are essential for both long-term applications and short-term applications (like precise rocket launches).

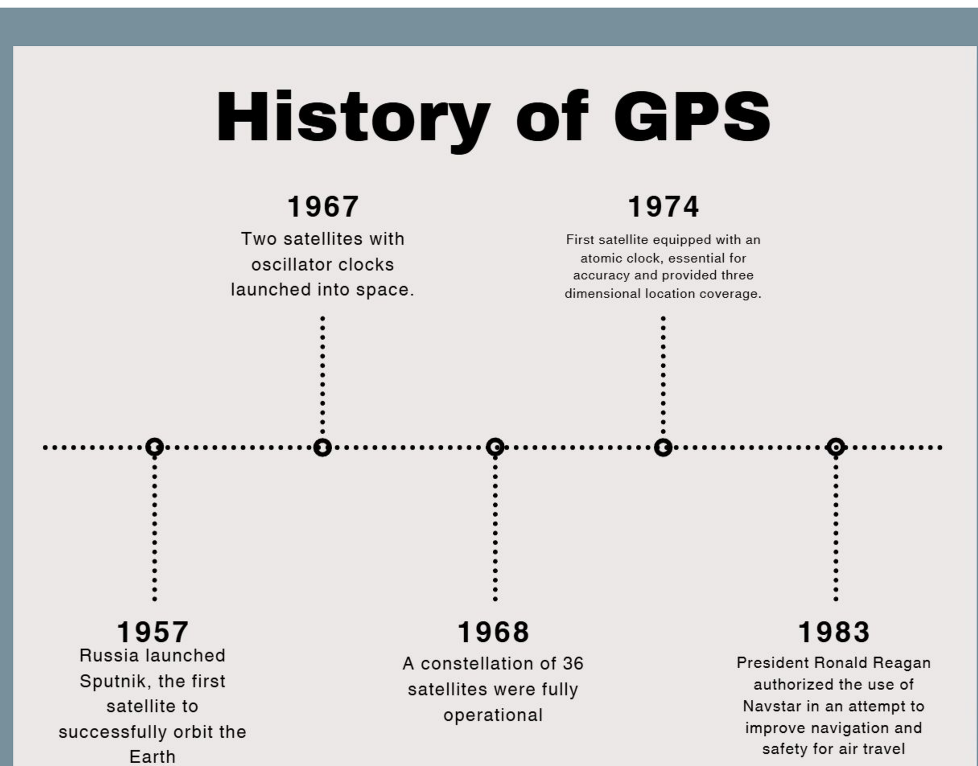


Figure 2: A brief history on the evolution of GPS that explains how GPS was used and improved over the years.

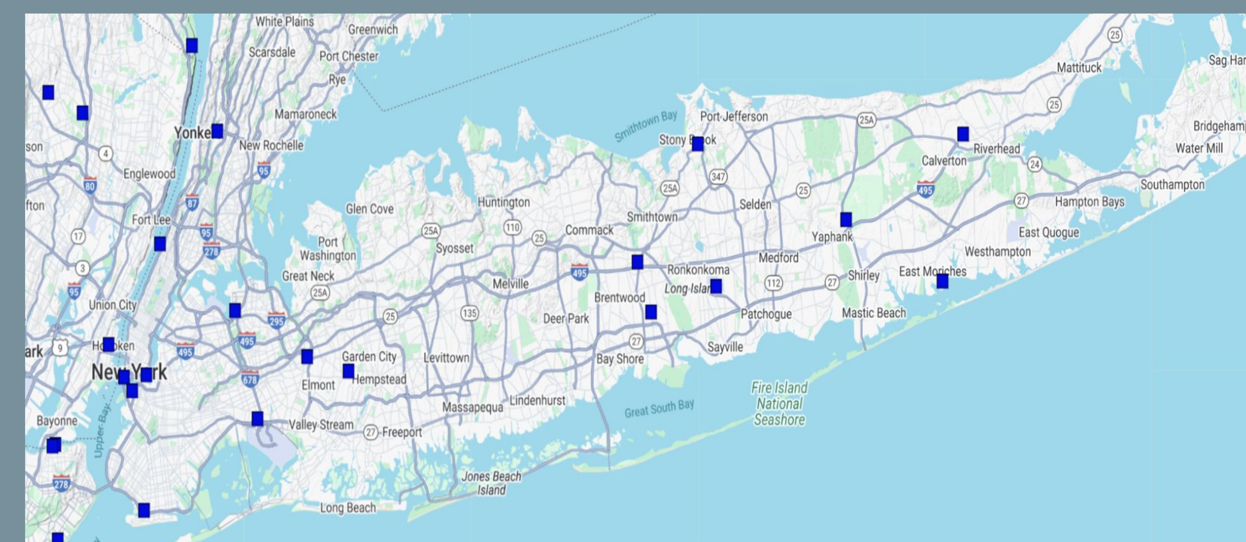
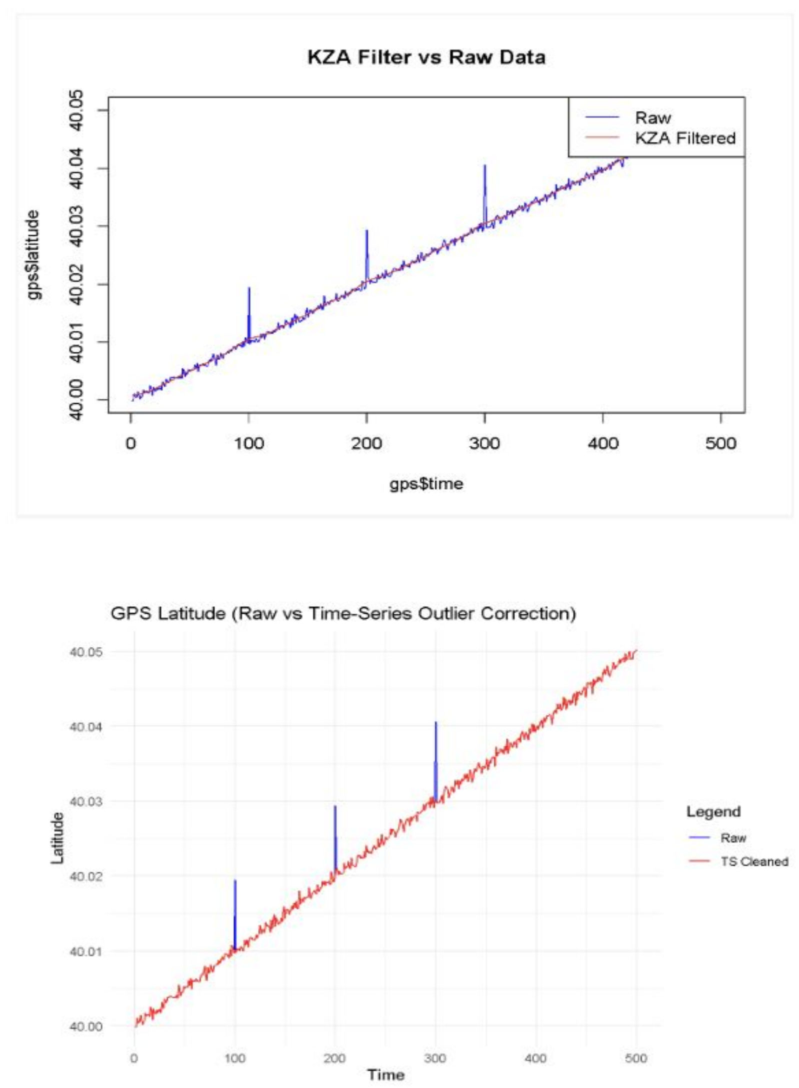


Figure 3: Map of Long Island's recorded GPS Stations. Blue dots represent the stations used to gather data of final orbits and rapid orbits. These sites are located in areas of higher elevation.



Outlier Detection using KZA + Replacement that is to remove values being far from a rolling mean, and replace the previously deleted values with the last or next values.

Outlier detection using the Time-Series method to model the underlying structure (trend + seasonality), so they can identify points that break the expected pattern.

Figure 4: Two methods for noise filtering and outlier correction to turn raw GPS into a clearer signal.

Credit Authorship Contribution Statement:

Wang J.: editing, literature, Rstudio coding, Figures, Tables, background, methods, results, discussion, conclusion, references, mapping, poster layout,abstract, teamwork and discussion with supervisor

Marsellos, A.E.: supervision, Rstudio coding, guidance, mentoring, tutoring.

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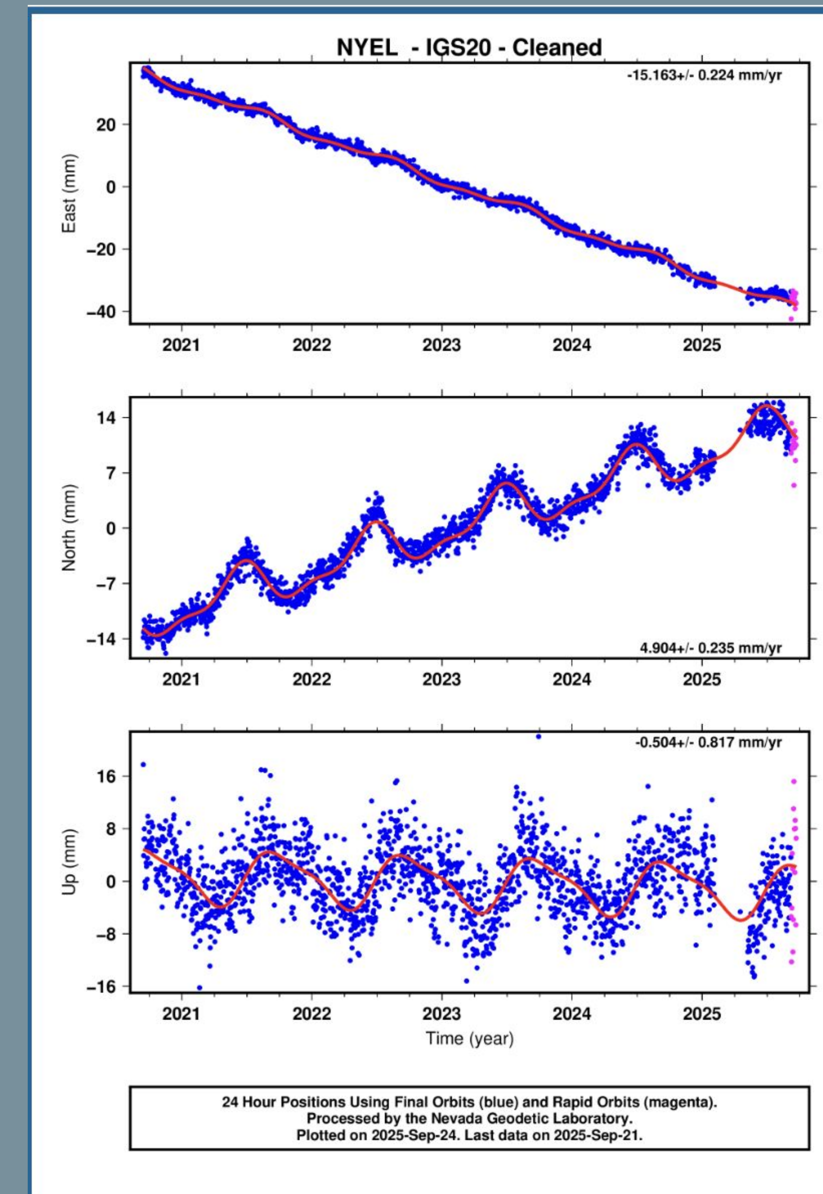

Figure 6: Processed GPS data (red) reduces noise and outliers in raw signals (blue), improving accuracy critical for rocket launch trajectories.

Located in Franklin Square, New York,

40°42'21.6"N 73°39'57.6"W

Discussion/Conclusion:

- Generating synthetic GPS data with noise
- Low-pass filtering (with cutoff guidance)
- Outlier removal with Z-scores (with trend caveat)
- Moving averages for smoothing (with edge effect caveat)

- Short-term → quick filtering, sensor fusion with IMU.
- Long-term → strong outlier removal, trend extraction.

- Fewer spikes/noise = Cleaner data and accuracy.

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Source (Google Slides):

https://docs.google.com/presentation/d/1eJMykDQTtEyK14bywY_vYtQWQhoZrW62Q4psFlQUCFA/edit

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